

Mustafa Aslan

PhD Researcher • Data Scientist

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EDUCATION

PhD in Probabilistic Modeling for Healthcare Management and Operations

Cardiff University, Cardiff, UK

Oct 2024 — Present

- Recipient of WGSSS-ESRC Studentship Award
- Project focused on enhancing discharge planning process using probabilistic modelling
 - Supervisors: [Prof Bahman Rostami-Tabar](#), [Dr. Jeremy Dixon](#)

MSc in Financial Mathematics

Middle East Technical University, Ankara, Turkey

Oct 2017 — Aug 2021

- [Dissertation](#): Effects of Exchange Rate Volatility and Firm-Specific Features on the Rates of Returns of the Manufacturing Firms Listed in Borsa İstanbul: A CAPM Approach
 - Statistical&Machine learning techniques used: Markov Switching GARCH Models, ARIMA, Panel Data Econometrics, Principal Component Analysis

BSc in Business Administration

Middle East Technical University, Ankara, Turkey

Oct 2011 — Aug 2015

PUBLICATIONS & WORKING PAPERS

Integrating Regime-Switching Probabilistic Forecasts into Stochastic Optimization for Nurse Rostering in Mental Health Wards

European Journal of Operational Research

2026

- *Status*: Revise & Resubmit (Major Revision)

TEACHING

Graduate Tutor in Business Data Analytics

Cardiff University, Cardiff, UK

Oct 2025 — Present

- Designed lab work materials and lesson plans for MSc-level courses on Business Data Analytics, utilizing Python and R programming.
- Conducted weekly lab sessions, guiding and assisting students through practical applications of data analysis techniques.

Training sessions for [Data Lab for Social Good Research Group](#) members

- Causal Inference via Machine Learning Feb 2026
- Introduction to Dynamic Programming and Reinforcement Learning May 2025
- Introduction to Python and Quarto on Visual Studio Code Mar 2025

WORK EXPERIENCES

Reporting and Data Analytics Executive

AKBANK (a leading bank in Turkey), Istanbul

Jun 2022 — Sep 2024

- Applied time-series forecasting and machine learning techniques, including ARIMA, Bayesian Time Series, Prophet, ANN, LSTM, Random Forest, LightGBM and XGBoost, to historical data for making long-horizon forecasts of daily customer call volume. The winning model helped to allocate resources effectively, reducing customer wait times by 6%.

- Implemented machine learning models (e.g., XGBoost, LightGBM, CatBoost) to predict customer behavior with an over recall rate of 70% and an accuracy rate of 85%, enabling fewer customer complaints and increase sales by 14%.
- Analyzing large amounts of data to identify trends and find patterns, signals and hidden stories within customer calls data.
- Applied machine learning techniques (Z-score, IsolationForest) to detect anomalies.
- Applied unsupervised machine learning techniques (DBSCAN, Gaussian mixture, K-means) to cluster customers.
- Hyperparameter tuning for machine learning models using Hyperopt, Optuna and KerasTuner.

Research Associate

The Economic Policy Research Foundation of Turkey (Think Tank), Ankara

Jan 2022 — Apr 2024

- Determined areas of research to increase knowledge in the particular field.
- Utilizing inferential statistics such as hypothesis testing (e.g., t-test, ANOVA test, population proportion test), confidence intervals, correlation analysis and regression analysis to make inferences and draw conclusions about data.
- Developed statistical models (regression analysis, panel data modeling) for regional development projects to contribute to data-driven decisions.

Senior Process Development Analyst

ETI GIDA Inc (a major FMCG player in Turkey), Eskisehir

Jun 2019 — Jan 2022

- Interacted with internal customers to understand business needs and translate into requirements and project scope.
- Assessed the impact of current business processes on users and stakeholders and evaluated potential areas for improvement.
- Maintained strong working knowledge of ERP (SAP), CRM and business intelligence tools and operational features.

Internal Auditor

Turk Telekom Inc. (the telecom giant of Turkey), Ankara

Nov 2015 — Jun 2019

- Performed strategic planning, execution and finalization of audits using data analytics and critical thinking skills.
- Investigated discrepancies discovered during the auditing process.
- Recommended new methods to improve internal controls and operating efficiency.

RESEARCH INTERESTS

- Healthcare analytics
- Data-driven decision making
- Statistical learning and Machine learning
- Reinforcement learning and stochastic optimization
- Time series forecasting
- Probabilistic forecasting

SKILLS & EXPERTISE

Expertise: Mathematical and Statistical Modeling, Machine Learning, Time Series Analysis and Forecasting, Stochastic Optimization and Reinforcement Learning, Statistical and Explanatory Data Analysis, web scraping in Python

Programming:  Python  SQL

Reporting:  Quarto  Advanced Excel  QlikView  SAS

Languages English (Fluent), Kurdish (Native), Turkish (Native)

PROFESSIONAL DEVELOPMENT

- Certification:
 - [Data Science: Machine Learning, HarvardX \(edX\)](#)
- Books (some of my go-to books):
 - Forecasting: Principles and Practice (Hyndman et al, 2021)
 - Introduction to Statistical Learning with Applications in R (Tibshirani et al, 2019)
 - The Elements of Statistical Learning (Tibshirani et al, 2008)
 - Probabilistic Machine Learning: An Introduction (Murphy, 2022)
 - Reinforcement Learning: An Introduction (Barto et al, 2018)
 - Reinforcement Learning and Stochastic Optimization (Powell, 2022)
 - A Student's Guide to Bayesian Statistics (Lambert, 2018)
 - Dive into deep learning (Zhang, 2022)

AWARDS & COMPETITIONS

Winner (1st Place, 1–5 Day Horizon) & 2nd Place (5–10 Day Horizon)

SPHERE-PPL Forecasting Contest — NHS Acute Patient Harm Forecasting

2026

- Member of team *Data Lab for Social Good* (led by Prof. Bahman Rostami-Tabar).
- Developed the winning forecasting model implemented in the forecasting package I have been developing, [peshbeen](#), predicting daily avoidable NHS hospital deaths from ED admission delays.
- Secured **1st place** (1–5 day horizon) and **2nd place** (5–10 day horizon, behind Microsoft Research).

Welsh Graduate School for the Social Sciences (WGSSS) Studentship Award

Aug 2024

NATCOR Bursaries / EURO Funding for Stochastic Modelling

Mar 2025

The Best Presentation Award at the 5th Welsh Postgraduate Research Cluster Workshop

Sep 2025

REFERENCES

Available upon request